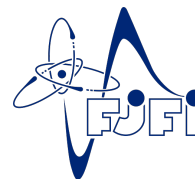




02YELMA Electricity and Magnetism



Lecturer: İskender Yalçinkaya, Ph.D.
Email: iskender.yalcinkaya@fjfi.cvut.cz
Web: <https://people.fjfi.cvut.cz/yakcuisk/index.html>
Office: B-509*
Office Hours: By appointment

*From the main staircase of the Břehová building, go up to the 4th floor, then take the stairs on your left toward the roof and ring the bell marked Q3.

Course Description

This summer semester course for first-year bachelor students covers the basics of electricity and magnetism, providing a solid foundation in these fundamental concepts, which are essential for understanding more advanced topics in physics and engineering.

Each week, there will be four 50-minute lectures and two 50-minute exercise session. Lectures will cover theoretical concepts following the weekly schedule, while exercise sessions will focus on solving problems related to the lecture material. The course is conducted in English.

Prerequisite(s)

Knowledge of basic differentiation, integration, and trigonometry would be useful but are not necessary.

Credits: 6

Reference Textbooks

1. Griffiths, D. J., *Introduction to Electrodynamics*, Cambridge University Press, 3rd Edition, 1999.
2. Purcell, E. M., and Morin D. J. *Electricity and Magnetism*, Cambridge University Press, 3rd Edition, 2013.
3. Pandey, D. C., *Electricity and Magnetism*, Arihant Publications, 16th Edition, 2020.

Course Objectives

1. To have a sound understanding of electricity and magnetism.
2. To be able to solve questions related to practical applications related to the subjects learned.

Homeworks

Each week, students will be given homework on the topics covered in class. The purpose of homework is that students can work on their own and familiarize themselves with the subject. Homework will not be returned and will not be graded. Solutions to the homework assignments will be provided one week before the upcoming exam date.

Exams

There will be **two midterm exams** and **one final exam**. Each exam will contain **four questions**, with each question worth **one point**. Students who obtain at least **50% of the total points** from the midterms will be assigned a Z grade and will be qualified for the final exam.

The final exam consists of a **written and an oral component**:

- Only students who score at least **50% on the written part** will proceed to the oral exam.
- The **final letter grade** is primarily determined by the written exam score. However:
 - If the oral exam score is **above 75%**, the final grade is increased by one level.
 - If the oral exam score is **below 25%**, the final grade is decreased by one level.
 - If the oral exam score is **between 25% and 75%**, the final grade remains the same.
- Attendance at the oral exam is **mandatory** for students who pass the written part.
- The structure of the oral part will be as follows: Each student will choose a random number from 1 to 16, corresponding to one of the sixteen topics listed at the end of this syllabus (please see “Topics for the Oral Part of the Final exam”). The students will then have 30 minutes to prepare a presentation on the chosen topic on a blank paper, followed by a 5 to 10-minute oral examination based on that topic. The students do not need to provide all the derivations (unless otherwise stated) but should be able to explain the main concepts and equations related to the topic.

The dates of the exams will be announced in the second week of the semester in KOS. The students are supposed to register for the exams in KOS. The students are required to bring their student ID cards to the exams.

Letter Grade Distribution

[3.6 – 4.0] : A = Excellent	[2.4 – 2.8) : D = Average
[3.2 – 3.6) : B = Very Good	[2.0 – 2.4) : E = Sufficient
[2.8 – 3.2) : C = Good	[0.0 – 2.0) : F = Insufficient

Makeup Exams

If a student fails to obtain at least 50% of the total points from the midterm exams required to qualify for the final exam, they are eligible to take a makeup exam. The makeup exam will be scheduled before the official examination period. The structure and grading of the makeup exam will be identical to the midterms and will cover all the topics from the first two midterms. After taking the makeup exam, the student must achieve at least 50% of the total points from the combined score of the midterms and the makeup exam in order to be eligible to attend the final exam.

Retaking the Course

If a student fails to obtain a Z grade for the first time, they must retake the course in the following academic year. The student is required to attend all classes again and retake all examinations. If the student fails this retake, they must obtain special permission from the study department to retake the course one final time.

If a student receives a Z grade but fails the final exam, the Z grade remains valid. In this case, the student is required to retake only the final exam in subsequent terms during examination period. If the student fails the final exam again, they must obtain special permission from the study department to retake the exam one final time.

Attendance and Absences

- Attendance is strongly recommended but not obligatory. There is a strong correlation between attendance and performance in the exams, so students are encouraged to attend all lectures and exercise sessions.
- Students are responsible for all missed work, regardless of the reason for absence. It is also the absentee's responsibility to get all missing notes or materials.

Tentative Course Outline

The weekly coverage might change as it depends on the progress of the class.

Week	Content
Week 1	Vectors, vector operations, gradient, divergence, curl, and their fundamental theorems. Line, surface, and volume integrals. Curvilinear coordinates, Dirac delta function, vector field theory, potentials, and Helmholtz theorem.
Week 2	Coulomb's law, superposition principle, electric fields from charge distributions, field lines, and Gauss's law.
Week 3	Electric potential, Poisson's/Laplace's equation, work and energy in electrostatics.
Week 4	Boundary conditions, conductors, insulators, Faraday cage, surface charges, force on conductors, and capacitors.
Week 5	Multipole expansion, dipoles, induced dipoles, polarization, bound charges, and their physical meaning.
Week 6	Lorentz force, cyclotron motion, current density, continuity equation, Biot-Savart law, and Ampère's law.
Week 7	Magnetic vector potential and its multipole expansion. Magnetic fields in matter, torque on magnetic dipoles, atomic orbits, magnetization, paramagnetism, diamagnetism.
Week 8	Field of a magnetised object, bound currents and their physical interpretation, an introduction to ferromagnetism.
Week 9	Conductivity, resistivity, Ohm's law, electrical circuits, electromotive force (emf), batteries, motional emf, and electromagnetic induction (Faraday's and Lenz's laws). Eddy currents.
Week 10	Inductance, RLC circuits, energy in magnetic fields, and Maxwell's equations.
Week 11	Wave equation, sinusoidal waves, electromagnetic waves in vacuum, monochromatic plane waves, and Poynting's theorem.
Week 12	Einstein's postulates, time dilation, Lorentz contraction, Galilean and Lorentz transformations, spacetime structure, proper time/velocity.
Week 13	Relativistic energy and momentum. Magnetism as a relativistic effect, field transformations, field tensor, and moving charge fields.

Topics for the Oral Part of the Final Exam

1. Explain Coulomb's law and the concept of the electric field. How do we go from the force between two point charges to the idea of a field that exists in space? Discuss the principle of superposition and how the electric field of a continuous charge distribution is calculated. Give examples.
2. State Gauss's law in both integral and (if you know) differential form. What is electric flux, physically? Explain why Gauss's law, while always true, is only useful for calculating electric fields in cases of high symmetry. Illustrate with at least two examples (e.g., infinite line charge, spherical shell, infinite plane).
3. Describe the properties of a conductor in electrostatic equilibrium (field inside, surface charge, potential). Explain why each property holds. What happens when a conductor is placed in an external electric field? Discuss the concept of electrostatic shielding and the Faraday cage.
4. Define electric potential and explain how it is related to the electric field (both as a line integral and as a gradient). What are equipotential surfaces, and why are they always perpendicular to field lines? Discuss the physical meaning of potential and potential difference, and how they differ from potential energy.
5. Explain what capacitance is and how it depends on geometry (give examples). How is energy stored in a capacitor? What happens physically when a dielectric material is inserted between the plates? Discuss the concepts of polarization and bound charges and how they modify the field.
6. Explain the microscopic picture of current flow in a conductor (drift velocity, collisions). What is the origin of electrical resistance? State Ohm's law and discuss its limitations — is it a fundamental law or an empirical relation? What determines the resistivity of a material?
7. State Kirchhoff's junction rule and loop rule. What fundamental conservation laws do they each express? Explain how these rules are applied to analyze a multi-loop DC circuit. Discuss the behavior of resistors in series and parallel and derive the combination rules.
8. Describe the magnetic force on a moving charged particle (Lorentz force). Why does the magnetic force do no work? Explain the motion of a charged particle in a uniform magnetic field. Extend the discussion to the force on a current-carrying wire and the torque on a current loop (magnetic dipole moment).
9. Explain how moving charges and currents create magnetic fields. State the Biot-Savart law and Ampère's law. In what sense is Ampère's law the magnetic analog of Gauss's law? Give examples where each is most useful (e.g., long straight wire, solenoid, toroid).
10. Explain why the magnetic field can be written as $\mathbf{B} = \nabla \times \mathbf{A}$. What role does $\nabla \cdot \mathbf{B} = 0$ play in guaranteeing this? Why isn't $\mathbf{A}$ unique — what is gauge freedom, and why doesn't it affect the physics? Compare the vector potential $\mathbf{A}$ to the scalar potential V in electrostatics. How is $\mathbf{A}$ calculated from a given current distribution? Discuss a concrete example.
11. State Faraday's law of electromagnetic induction. What is an electromotive force (EMF), and how does a changing magnetic flux produce one? State Lenz's law and explain how it is a consequence of conservation of energy. Give at least two concrete examples (e.g., moving bar on rails, rotating loop).

12. Define self-inductance and mutual inductance. Derive or explain the self-inductance of a solenoid. How does an inductor behave in a circuit, and why does it oppose changes in current? Discuss the energy stored in an inductor and the concept of energy density in a magnetic field. Compare with the analogous capacitor/electric field case.
13. Describe what happens when a capacitor charges/discharges through a resistor (RC circuit) and when current builds up/decays in an RL circuit. What is the time constant in each case, and what does it represent physically? Discuss the energy considerations — where does the energy go during these transient processes?
14. Describe what happens when a capacitor charges/discharges through a resistor (RC circuit) and when current builds up/decays in an RL circuit. What is the time constant in each case, and what does it represent physically? Discuss the energy considerations — where does the energy go during these transient processes?
15. Explain how Maxwell's equations predict the existence of electromagnetic waves. Describe the key properties of an electromagnetic wave (transverse nature, relationship between E and B , speed, energy transport via Poynting vector). Why was it significant that the predicted speed matched the measured speed of light?
16. Explain the idea of expanding the potential of a localized charge distribution in powers of $1/r$. What is the physical meaning of the monopole and dipole terms? Define the electric dipole moment and describe the potential and field of a pure dipole. How does a dipole behave in an external field?
17. Why is there no magnetic monopole term in the multipole expansion for magnetism? Define the magnetic dipole moment of a current loop and describe the field it produces. Compare the fields of a magnetic dipole and an electric dipole — what is similar and what differs?
18. State Poynting's theorem and explain the physical meaning of each term. Define the Poynting vector and its interpretation. Give a concrete example (e.g., a resistive wire carrying steady current or a charging capacitor) and discuss how energy flows through fields.
19. Explain what happens microscopically when a dielectric is placed in an external electric field. Define polarization P and show how it gives rise to bound surface charge density σ_b and bound volume charge density ρ_b . How do these bound charges modify the total electric field? Illustrate with a concrete example such as a uniformly polarized sphere or a polarized slab.